THE RIGHT ADJOINT Q OF THE INCLUSION QCOH INTO LQCOH^{FBC}

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author omits the routine verification of the cocycle condition for the descent datum alpha; the construction itself is complete.

Allowed standard prerequisites (editorial).

- Stack presentation by a smooth groupoid in spaces. Algebraic Stacks, Lemma algebraic-lemma-stack-presentation; Sheaves on Stacks, Proposition stacks-sheaves-proposition-quasi-coherent. Presentation and equivalence.
- Groupoid modules descent datum. Groupoids in Spaces, Definition spaces-groupoids-definition-groupoid-module. Cocycle condition (omitted routine check).

Source reference. Original section 0GQG: sources/stacks-chow/tags.txt, source lines 20861–20861. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04T5: sources/stacks-chow/tags.txt, source lines 6243–6243. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WT: sources/stacks-chow/tags.txt, source lines 8827–8827. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0441: sources/stacks-chow/tags.txt, source lines 5384–5384. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 076S: sources/stacks-chow/tags.txt, source lines 9177–9177. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Let $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ be the category of locally quasi-coherent modules with the flat base change property, see Section 0GQG. The inclusion functor $i : QCoh(\mathcal{O}_\mathcal{X}) \to LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ has a right adjoint*

$$Q : LQCoh^{fbc}(\mathcal{O}_\mathcal{X}) \to QCoh(\mathcal{O}_\mathcal{X})$$

such that $Q \circ i$ is the identity functor.

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Proof. Choose a scheme U and a surjective smooth morphism $f : U \to \mathcal{X}$. Set $R = U \times_{\mathcal{X}} U$ so that we obtain a smooth groupoid (U, R, s, t, c) in algebraic spaces with the property that $\mathcal{X} = [U/R]$, see Algebraic Stacks, Lemma 04T5. We may and do replace $\mathcal{X}$ by $[U/R]$. By Sheaves on Stacks, Proposition 06WT there is an equivalence

$$q_1 : QCoh(U, R, s, t, c) \longrightarrow QCoh(\mathcal{O}_{\mathcal{X}})$$

Let us construct a functor

$$q_2 : LQCoh^{fbc}(\mathcal{O}_{\mathcal{X}}) \longrightarrow QCoh(U, R, s, t, c)$$

by the following rule: if $\mathcal{F}$ is an object of $LQCoh^{fbc}(\mathcal{O}_{\mathcal{X}})$ then we set

$$q_2(\mathcal{F}) = (f^*\mathcal{F}|_{U_{étale}}, \alpha)$$

where α is the isomorphism

$$t^*_{small}(f^*\mathcal{F}|_{U_{étale}}) \to t^*f^*\mathcal{F}|_{R_{étale}} \to s^*f^*\mathcal{F}|_{R_{étale}} \to s^*_{small}(f^*\mathcal{F}|_{U_{étale}})$$

where the outer two morphisms are the comparison maps. Note that $q_2(\mathcal{F})$ is quasi-coherent precisely because $\mathcal{F}$ is locally quasi-coherent and that we used (and needed) the flat base change property in the construction of the descent datum α. We omit the verification that the cocycle condition (see Groupoids in Spaces, Definition 0441) holds. Looking at the proof of Sheaves on Stacks, Proposition 06WT we see that $q_2 \circ i$ is the quasi-inverse to q_1. We define $Q = q_1 \circ q_2$. Let $\mathcal{F}$ be an object of $LQCoh^{fbc}(\mathcal{O}_{\mathcal{X}})$ and let $\mathcal{G}$ be an object of $QCoh(\mathcal{O}_{\mathcal{X}})$. We have

$$\begin{aligned} \mathrm{Mor}_{LQCoh^{fbc}(\mathcal{O}_{\mathcal{X}})}(i(\mathcal{G}), \mathcal{F}) &= \mathrm{Mor}_{QCoh(U,R,s,t,c)}(q_2(i(\mathcal{G})), q_2(\mathcal{F})) \\ &= \mathrm{Mor}_{QCoh(\mathcal{O}_{\mathcal{X}})}(\mathcal{G}, Q(\mathcal{F})) \end{aligned}$$

where the first equality is Sheaves on Stacks, Lemma 076S and the second equality holds because $q_1 \circ i$ and q_2 are quasi-inverse equivalences of categories. The assertion $Q \circ i \cong \mathrm{id}$ is a formal consequence of the fact that i is fully faithful. □